Implementation plan — AwkBeats Prep Station

Document control

Field	Value
Project	AwkBeats Prep Station
Document title	Implementation Plan
Document ID	ABP-PLAN
Version	1.0
Status	Final (as shipped)
Last updated	2026-10-21 (UTC)
Author	Cursor Agent, for owner Awkward J
Repo	https://github.com/Awkward8787/awkbeats-prep-station
Branch	`cursor/awkbeats-prep-station-e2e2`
Commit SHA	f99d3c6a38ad256e123fa13b50e0f7660369bd61
Live URL	None. Installers are GitHub Actions artifacts, not a Release.
Operator	Awkward J (legal entity UNCONFIRMED)

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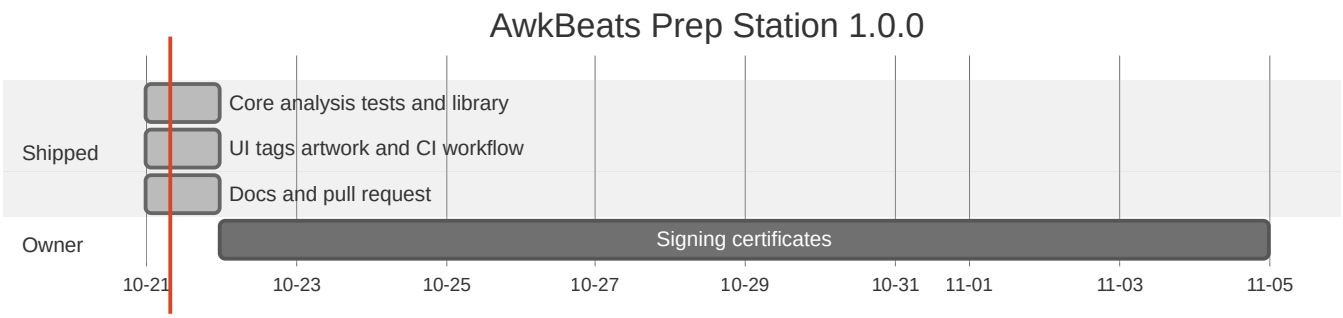
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1. Delivery summary

AwkBeats Prep Station 1.0.0 was built in this repository as a local Electron app: library, offline BPM and key analysis, tag writing, cover art, CSV, batch tools, tests, and a GitHub Actions workflow that builds an unsigned macOS dmg and a Windows exe. No GitHub Release is published. The builder was Cursor Agent. The owner is Awkward J. Status: shipped as source and as CI installer artifacts once the shipping commit's workflow is green.

2. Phases and milestones

Dates are the day the tree was produced (2026-10-21). They are not estimates of future work.



The build landed in one working tree. The signing bar is the owner's, and the two weeks on the chart are a placeholder span, not a promise.

Phase	Status	Result
Analysis, library, tags, tests	Shipped	`src/core`, `tests/`
UI and installers workflow	Shipped	`src/renderer`, `.github/workflows/installers.yml`
Documentation and PDFs	Shipped	`docs/`
Signed release	Planned	Blocked on certificates. Not started.

3. Work breakdown

ID	Task	Phase	Requirements	Status	Notes
TASK-001	Original BPM and key detectors, no GPL library	Build	FR-004, FR-005, NFR-001	Shipped	`src/core/analysis`
TASK-002	Reference-tone tests at 0.01 BPM and 24 keys	Build	NFR-002	Shipped	`tests/bpm-key.test.ts`
TASK-003	Library JSON, import, filters, CSV, locks	Build	FR-001–FR-003, FR-006, FR-011, FR-012	Shipped	`src/core/library`
TASK-004	ID3, RIFF INFO, Vorbis, artwork embed	Build	FR-008, FR-009	Shipped	`TagWriter.ts`
TASK-005	Dark UI, batch, crop, shortcuts	Build	FR-007, FR-013–FR-017, NFR-004	Shipped	`src/renderer`
TASK-006	License check in CI	Build	NFR-001	Shipped	`scripts/check-licenses.mjs`
TASK-007	Actions workflow for dmg and exe	Build	NFR-003	Shipped	artifacts, `--publish never`
TASK-008	Contrast fix for faint text	Build	NFR-008	Shipped	`--text-faint #938d85`
TASK-009	Six docs and PDFs	Docs	—	Shipped	this folder
TASK-010	Apple notarization	Owner	NFR-003 follow-up	Planned	Secrets in README
TASK-011	Windows Authenticode	Owner	NFR-003 follow-up	Planned	EV recommended
TASK-012	Confirm legal entity and license	Owner	Q-001, Q-002	Planned	LICENSE is proprietary pending confirmation
TASK-013	Brand approval of the AJ mark	Owner	Q-004	Planned	`build/icon.svg`

ID	Task	Phase	Requirements	Status	Notes
TASK-014	Do not publish a Release	Build	brief	Shipped	`publish: null`

The pull request that contains this file is the delivery PR into `main`. Commit SHAs are on that branch.

4. Review gates

Review	Date	Verdict	Findings	Resolution
Unit tests	2026-10-21	Pass required before commit	BPM and tag assertions kept strict	Shipped with the tests green
License scan	2026-10-21	Pass	Production tree had no GPL-family license	`npm run licenses`
Typecheck	2026-10-21	Pass	`tsc --noEmit`	Shipped
Contrast	2026-10-21	Fail, then pass	`#7e786e` was 4.02:1 on elevated surfaces	Changed to `#938d85`, minimum 4.50:1
External code review, accessibility audit, security audit	—	Not run	No third-party report was attached to the brief	Deferred. Debt is listed in ABP-TRD.
UI check on Electron	2026-10-21	Exercised	Library, analysis card, metadata, crop dialog captured	Screenshots are PR artifacts

No finding was waived by weakening a test.

5. Testing and QA

Automated: `npm test`, `npm run typecheck`, `npm run licenses`. What each test file covers is listed in ABP-TRD section 13.

Manual script used on the Linux Electron build:

1. Launch with `AWKBEATS_USER_DATA` pointed at a scratch directory.
2. Import four generated loops (WAV). Confirm the table lists them.
3. Select the 140.25 BPM F# minor loop. Confirm the inspector shows 140.25, F# minor, and a Camelot badge.
4. Scroll to cover, title, producer, genre, mood, and tags.
5. Open Adjust crop. Confirm the export size control offers 3000×3000.
6. Half-time and double-time are unit-tested (`BpmMath.half(140.25)` is 70.13, and the scale is always applied to the detected tempo). The buttons call `save({ bpmScale: 0.5 | 2, bpmOverride: null })`.

Devices: development GUI on Linux with Electron 33.4.11. macOS and Windows are covered by the installer jobs compiling, not by a human clicking the packaged app in this session. Browsers: not applicable.

Results: unit tests and typecheck passed on the shipping tree. Installer artifact names are confirmed when the Actions run for this commit is green. If that run fails, the fix lands on the same branch before the PR is treated as done.

6. Launch checklist

Item	State	Evidence
Domain and DNS for the app	Not applicable	No app host. awkbeats.com is the brand site and is not changed by this repo.
SSL for the app	Not applicable	Local app
Secrets for unsigned build	Done	None required

Item	State	Evidence
Signing secrets	Not done	Owner
Payments	Not applicable	No provider
Legal pages inside the app	Not done	Not in scope for v1. Sale terms are Q-005.
Analytics	Done as none	No SDK
CSP	Done	`src/renderer/index.html`
404 page	Not applicable	No router
App icon	Done	`build/icon.png`, `build/icon.ico`
Performance	Done as measured	ABP-TRD section 9
Accessibility contrast	Done	ABP-DESIGN section 3
Backups	Not done by the app	Producer copies userData
Monitoring	Not applicable	No service
Installers	Workflow shipped	Artifact upload steps
Release published	Not done, on purpose	`--publish never`

7. Deployment runbook

There is no server deploy.

To produce installers from a desktop:

1. Install Node 22.
2. `npm ci`
3. On a Mac: `npm run dist:mac`. On Windows: `npm run dist:win`.
4. Take the files in `release/`. Do not run electron-builder publish.

From GitHub:

1. Push the branch. Workflow `Build installers` starts on every push.
2. Open the run. Download `AwkBeats-Prep-Station-macOS` and `AwkBeats-Prep-Station-Windows`.
3. Expected names: `AwkBeats-Prep-Station-1.0.0-mac-arm64.dmg`, `AwkBeats-Prep-Station-1.0.0-mac-x64.dmg`, `AwkBeats-Prep-Station-1.0.0-win-x64.exe`.

Verify: tests job green, both artifact jobs green, files non-empty. Rollback: download the previous run's artifacts. Access: whoever can push to the repo can start a build. Signing secrets, once added, should be environment secrets limited to maintainers.

Do not create a GitHub Release from this workflow.

8. Post-launch and owner actions

No temporary files need deleting in the repo. `dist/` and `release/` are gitignored.

Owner actions, in order:

1. **Apple signing and notarization.** Enroll the legal entity in the Apple Developer Program. Create a Developer ID Application certificate. Export the `.p12`. Create an App Store Connect API key. Add GitHub secrets `CSC_LINK` (base64 p12), `CSC_KEY_PASSWORD`, `APPLE_API_KEY` (base64 p8), `APPLE_API_KEY_ID`, `APPLE_API_ISSUER`,

`APPLE_TEAM_ID`. Then set `mac.identity`, `hardenedRuntime: true`, and notarization in `electron-builder.yml`. Details are in the README.

2. **Windows code signing.** Buy an OV or EV certificate. EV is what SmartScreen trusts without a reputation warmup. Add `WIN_CSC_LINK` and `WIN_CSC_KEY_PASSWORD` (or `CSC_LINK` / `CSC_KEY_PASSWORD`). `signAndEditExecutable` is already true.
3. **Confirm the legal entity** for the copyright string. The app currently says Awkward J. Q-001.
4. **Confirm the license.** The repo ships `LICENSE` as all rights reserved because the app may be sold. Say if that should change. Q-002.
5. **ISRC.** The check is 12 characters. Provide a registrant prefix only if you want a tighter check. Q-003.
6. **Brand.** Approve `build/icon.svg` or send a replacement. Q-004.
7. **Distribution.** Decide where installers are downloaded. Do not use a GitHub Release until you ask for one. This delivery does not publish one.
8. **Support email.** None is shown. Send one if it should appear. Q-006.

9. Roadmap and next phases

ID	Task	Requirements	Depends on	Status	Size
TASK-010	Signed macOS build	NFR-003	Q-001, Apple secrets	Planned	One workflow change after secrets exist
TASK-011	Signed Windows build	NFR-003	Certificate	Planned	Same
TASK-015	Relink missing files	FR-020	Q-012	Planned	Small dialog
TASK-016	Optional image watermark	FR-010	Owner asks	Deferred	Not in v1
TASK-017	SQLite catalog	—	Catalog growth	Deferred	New store behind `LibraryStore`
TASK-018	Linux package	—	Owner asks	Deferred	electron-builder linux target
TASK-019	Focus trap on dialogs	NFR-007	—	Planned	`dialogs.tsx`
TASK-020	Price and storefront	Monetization	Q-005	Deferred	No number was given. Do not invent one.

"Size" is structural, not a schedule.

10. Risks and mitigations

Risk	Mitigation
CI cannot sign, so the first artifacts warn on open	Documented. Unsigned is the accepted v1 state.
A future contributor adds essentia or another GPL detector	`npm run licenses` fails the test job
Detector wrong on a dense mix	Override, lock, half-time, double-time. Tests stay strict on generated material.
`GITHUB_TOKEN` tempts electron-builder to publish	`publish: null` and `--publish never`

11. Maintenance plan

- Dependencies: review npm updates when touching the app. Re-run tests and `npm run licenses` after any audio or tag dependency bump. Do not upgrade music-metadata past a major version without re-checking decimal BPM, because 7.14.0's common reader drops the fraction and `TagReader.readNative` compensates.
- Secrets: only exist once the owner adds them. Rotate by replacing the GitHub secret and revoking the old certificate or API key at the vendor. No rotation is due today.

- Certificates: Apple and Windows certs expire on the vendor's calendar. Renew before the signed job starts failing.
- Content: genre and mood lists are in `BeatCatalog`. UI strings are in `UserMessage.ts` and the components.
- Cadence: re-run the installer workflow on each push. No separate staging cron.
- Docs: bump the document version and regenerate `docs/pdf` when behavior changes.

12. Glossary

Term	Meaning
Artifact	A file uploaded by GitHub Actions, downloaded from the workflow run. Not a GitHub Release.
Developer ID	Apple certificate for apps distributed outside the Mac App Store.
EV certificate	Extended Validation code-signing certificate for Windows.

13. Open questions

The open questions from the other documents apply here. The ones that block owner work:

ID	Question	Owner	Blocked
Q-001	Legal entity	Awkward J	Apple certificate name
Q-002	Keep the proprietary license?	Awkward J	LICENSE
Q-003	ISRC prefix	Awkward J	Tighter validation
Q-004	Approve the mark	Awkward J	Icon
Q-005	Sell or not, price, store	Awkward J	Monetization. No price in the brief.
Q-006	Support email	Awkward J	Footer or About
Q-007	OV or EV Windows certificate	Awkward J	SmartScreen
Q-009	Should remove ever delete audio?	Awkward J	A second confirm. Default remains no.

14. Changelog

Date	Version	Change	Why
2026-10-21	1.0	Plan for the 1.0.0 delivery, including owner signing steps and an explicit decision not to publish a Release.	Owner build brief.

15. Sources

- `.github/workflows/installers.yml`, `electron-builder.yml`, `README.md`, `LICENSE`.
- `package.json` scripts `dist:mac`, `dist:win`, `licenses`, `test`.
- ABP-PRD, ABP-TRD, ABP-FLOW, ABP-DESIGN, ABP-DATA in this folder.
- Owner brief: desktop app, permissive analysis license, dmg and exe artifacts, no release publish.